

Bugs found: 4

1. `NullPointerException` occurs because `s` is null when `s.length()` is called.
2. `ClassCastException` occurs because `o` references a `String` but is cast to `Integer`.
3. `ArrayIndexOutOfBoundsException` occurs because `arr[10]` accesses index 10 in an array of length 5.
4. Potential memory leak occurs because `cache` is static and retains added objects indefinitely.